



Faculty	FET		
Assessment Name	ICA3	Paper Code	CNH2/I26
Module Name	Managing Network Hardware 2	Module Code	C8-NH2-11
Semester	Jan-Jun 2026	Submission Mode	Blackboard
Total Marks	100	Duration	
Assessment Type	Practical	Submission Date	18/04/2026 11:00 AM

Instructions

- a) Submit packet tracer file on blackboard.
- b) The use of AI is not allowed during this assessment.

Background

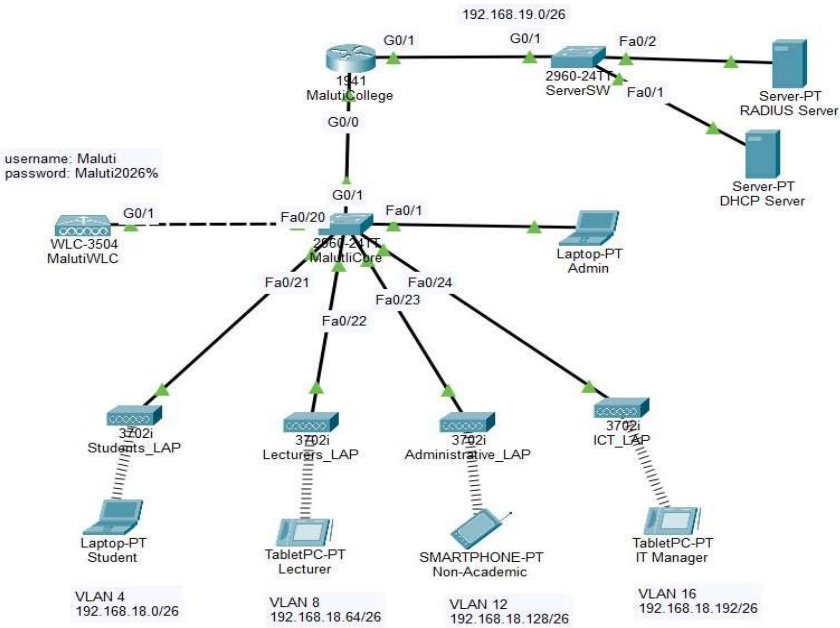
You have recently been appointed as a Network Administrator at Maluti College of Computing. The college needs a wireless network to support students, lecturers and administrative staff across campus buildings.

The network must:

- Provide secure, department-based wireless access.
- Allow communication between the departments.
- Be managed centrally through a Wireless LAN Controller (WLC).
- Authenticate users through a RADIUS server.
- Assign IP addresses dynamically through a DHCP server.

1. Network topology

a) Create the network topology and rename the devices as shown below. (5)



2. Admin PC

a) This PC must be used to manage the WLC. It must be assigned the third usable IP address in the ICT subnet. (2)

3. DHCP Server

- a) Assign the second usable IP address to the DHCP server in the ServerRoom subnet. (2)
- b) Create DHCP pools for the following VLANs: VLAN4, VLAN8, VLAN12 and VLAN16. Each VLAN must support at least 45 wireless users. (8)
- c) Exclude the first five IP addresses in each pool for network devices. (4)

d)

4. RADIUS Server

- a) Assign the third usable IP address to the RADIUS server in the ServerRoom subnet. (2)
- b) Configure RADIUS server for authentication. (secret: **Maluti2026%**) (2)
- c) Create the following user credentials to authenticate each user. (4)

Username	Password
student.student	student@123
lecturer.lect	lecturer@123
non.academic	nonacademic@123
manager.manager	manager@123

5. Router configurations

- a) Configure the device name as per the topology. (1)
- b) Configure MalutiCollege to enable communication between VLANs. Assign the first usable IP address in each VLAN as default gateway. (4)
- c) Configure ICT VLAN as a native. (1)
- d) Configure the router as a DHCP relay to forward broadcast messages to the DHCP server. (4)
- e) Assign the first usable IP address to the router interface that connects to ServerRoom subnet. (1)

6. MalutiCore configurations

- a) Configure the device names as per the network topology. (1)
- b) Create the VLANs as shown in the network topology. (4)
- c) Admin PC must be able to access VLAN 16. Assign its switch port to the appropriate VLAN. (1)
- d) Configure the trunk links as necessary. Use VLAN 35 to allow untagged traffic to pass through. (12)
- e) Only VLANs in the topology should be allowed on the trunk links. (6)
- f) Configure the ServerSW name as shown in the topology. (1)

7. Lightweight Access Points

- a) Must receive IP addresses from DHCP server and be managed through WLC. (4)

8. WLC Configurations

- a) Assigned the second usable IP address in the ICT subnet/VLAN to allow communication with other devices on the network. (2)
- b) An administrator must use the following credentials to access the WLC: (1)
 - Username: **Maluti**
 - Password: **Maluti2026%**
- c) System name: **MalutiWLC** (1)
- d) The Wireless LAN Controller must be configured with the correct IP address and be in the correct VLAN. (2)
- e) Create a shared secret key for RADIUS server as **Maluti2026%** (1)
- f) Create VLANs as per the topology and assign the second usable IP address in respective VLANs (9)
- g) Create WLANs for each department and map them to their respective VLANs. All the WI-FI clients must be authenticated through RADIUS server as per the table below. (7)

Profile	WLAN
ICT	ICT_WI-FI
Students	Students_WI-FI
Lecturers	Lecturers_WI-FI
Administrative	Administrative_WI-FI

- h) Restrict each SSIDs so that only users in the corresponding VLAN can connect. (4)

Group name	Description
Students	Connect only Students
Lecturers	Connect only Lecturers
+Administrative	Connect only non-academic staff
ICT-Group	Connect only ICT staff

9. Connectivity

- a) All wireless devices must obtain the correct IP addresses dynamically from the DHCP server and connect in their respective VLANs. (4)

